

Insider Threat Investigation Simulator Using AI-Assisted Behavioral Analysis

The goal of this document is to create a simple insider threat investigation simulator. We created sample security logs, combined them into one evidence file, and used AI tools to analyze suspicious activity and generate an incident report.

Part 1: Create Sample Insider Threat Logs

Create 3-4 simulated log files showing things like:

- Employee logging in after normal business hours
- Multiple failed logins followed by a successful login
- PowerShell execution
- Unauthorized file access

Run these in Kali inside your project folder:

```
mkdir -p ~/insider_threat_poc
```

```
cd ~/insider_threat_poc
```



```
File Actions Edit View Help
(labuser@kali)~[~]
$ mkdir -p ~/insider_threat_poc

(labuser@kali)~[~]
$ cd ~/insider_threat_poc
```

Step 1. Create simulated login log

```
cat > login_activity.log << 'EOF'
```

```
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
EOF
```

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```
(labuser@kali)-[~/insider_threat_poc]
$ cat > login_activity.log << 'EOF'
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
EOF
```

Step 2. Create suspicious file access log

```
cat > file_access.log << 'EOF'
```

```
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx action=READ
2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ
2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
EOF
```

```
(labuser@kali)-[~/insider_threat_poc]
$ cat > file_access.log << 'EOF'
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx action=READ
2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ
2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
EOF
```

Step 3. Create suspicious PowerShell log

```
cat > powershell_activity.log << 'EOF'
```

```
2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
ParentProcess=explorer.exe
Alert=Suspicious PowerShell execution outside business hours
EOF
```

```
(labuser@kali)-[~/insider_threat_poc]
$ cat > powershell_activity.log << 'EOF'
2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
ParentProcess=explorer.exe
Alert=Suspicious PowerShell execution outside business hours
EOF
```

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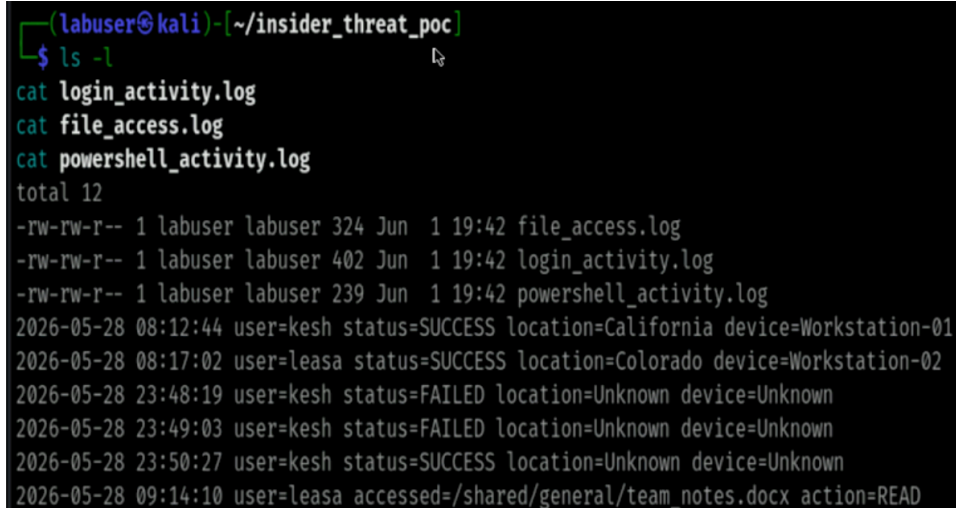
Step 4: Verify files were created

ls -l

cat login_activity.log

cat file_access.log

cat powershell_activity.log



A terminal window showing the execution of several commands to verify the existence and content of log files. The commands are: `ls -l`, `cat login_activity.log`, `cat file_access.log`, and `cat powershell_activity.log`. The output of `ls -l` shows three files: `file_access.log` (324 bytes), `login_activity.log` (402 bytes), and `powershell_activity.log` (239 bytes). The output of the `cat` commands shows a list of log entries, including user login attempts, status reports, and file access records.

```
(labuser@kali)~[~/insider_threat_poc]
$ ls -l
cat login_activity.log
cat file_access.log
cat powershell_activity.log
total 12
-rw-rw-r-- 1 labuser labuser 324 Jun  1 19:42 file_access.log
-rw-rw-r-- 1 labuser labuser 402 Jun  1 19:42 login_activity.log
-rw-rw-r-- 1 labuser labuser 239 Jun  1 19:42 powershell_activity.log
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx action=READ
```

Step 5: Combine everything into one evidence file

Run in Kali to create txt file for all logs together:

cat login_activity.log file_access.log powershell_activity.log > insider_threat_evidence.txt



A terminal window showing the execution of the command `cat login_activity.log file_access.log powershell_activity.log > insider_threat_evidence.txt`. The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The prompt shows the user is in the directory `~/insider_threat_poc`.

```
File Actions Edit View Help
(labuser@kali)~[~/insider_threat_poc]
$ cat login_activity.log file_access.log powershell_activity.log > insider_threat_evidence.txt
```

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Step 6 :Confirm file created

ls -l

cat insider_threat_evidence.txt

```
(labuser@kali)~[~/insider_threat_poc]
$ ls -l
cat insider_threat_evidence.txt
total 16
-rw-rw-r-- 1 labuser labuser 324 Jun  1 19:42 file_access.log
-rw-rw-r-- 1 labuser labuser 965 Jun  1 19:44 insider_threat_evidence.txt
-rw-rw-r-- 1 labuser labuser 402 Jun  1 19:42 login_activity.log
-rw-rw-r-- 1 labuser labuser 239 Jun  1 19:42 powershell_activity.log
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx action=READ
2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ
2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
ParentProcess=explorer.exe
Alert=Suspicious PowerShell execution outside business hours
```

You now have a consolidated evidence file named **insider_threat_evidence.txt** that can be analyzed using ChatGPT and TinyLLaMA to generate incident reports and identify suspicious activity.

Part 2: Use Chat GPT AI to review logs

Take files created and enter in this in chat GPT:

You are a SOC analyst.

Analyze the following insider threat evidence and provide:

1. Incident Summary
2. Timeline of Events
3. Suspicious Activity Identified
4. Risk Level (Low, Medium, High)
5. Recommended Actions

Logs:

here are the logs ls -l
cat insider_threat_evidence.txt

total 16

```
-rw-rw-r-- 1 labuser labuser 324 Jun  1 19:42 file_access.log
-rw-rw-r-- 1 labuser labuser 965 Jun  1 19:44 insider_threat_evidence.txt
-rw-rw-r-- 1 labuser labuser 402 Jun  1 19:42 login_activity.log
-rw-rw-r-- 1 labuser labuser 239 Jun  1 19:42 powershell_activity.log
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California
device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado
device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx
action=READ
2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ
2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
ParentProcess=explorer.exe
Alert=Suspicious PowerShell execution outside business hours
```

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Here are our findings for our report from Chat GPT

This is the type of report I would expect from your MVP. You could even use something very close to this as your `incident_report.txt`.

Incident Summary

A review of the security logs identified several suspicious activities involving the user **kesh**. The activity included multiple failed login attempts from an unknown location followed by a successful login, access to sensitive company files outside normal business hours, and execution of PowerShell using suspicious parameters. These events suggest potential insider threat activity or possible account compromise and warrant further investigation.

Timeline of Events

Timestamp	Event
2026-05-28 08:12:44	User kesh successfully logged into Workstation-01 from California
2026-05-28 08:17:02	User leasa successfully logged into Workstation-02 from Colorado
2026-05-28 09:14:10	

Part 3: Install Ollama and TinyLlama

Step 1 Update system by running these command before installation:

Command 1: `sudo apt update`

```
(labuser@kali) - [~/insider_threat_poc]
$ sudo apt upgrade -y

The following packages were automatically installed and are no longer required:
  cpp-13          libgpgme11t64    libpython3.11-minimal  node-cjs-module-lexer
```

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Command 2: sudo apt upgrade -y

```
(labuser@kali)~[~/insider_threat_poc]
$ sudo apt upgrade -y

The following packages were automatically installed and are no longer required:
  cpp-13                                libgpgme11t64                libpython3.11-minimal        node-cjs-module-lexer
  cpp-13-x86-64-linux-gnu               libgpgmepp6t64              libpython3.11-stdlib        node-corepack
  cpp-14                                libgspell-1-2               libqt5ct-common1.8         node-debug
  cpp-14-x86-64-linux-gnu               libgtk2.0-0t64              libroc0.3                   node-llhttp
  enchant-2                             libgtk2.0-bin               librubberband2              node-lru-cache
  firmware-ti-connectivity              libgtk2.0-common            libSDL2-2.0-0               node-minimatch
  gcc-13-base                           libgtkmm-3.0-1t64           libSDL2-classic             node-ms
  gir1.2-girepository-2.0               libical3t64                 libsframe2                  node-semver
  ibus-gtk                              libicu76                    libsigc++-2.0-0v5          node-undici
  libabsl20230802                       libinstpatch-1.0-2         libsimdutf27                node-xtend
  libapt-pkg6.0t64                     libjim0.82t64               libsnmp40t64                nodejs
  libassuan0                            libldap-2.5-0               libsodium23                  nodejs-doc
  libatkmm-1.6-1v5                     libllhttp9.3                libspnbase3t64              openjdk-21-jre
  libaudio2                             libllvm17t64                libswscale8                  openjdk-21-jre-headless
  libavc1394-0                          libmbc16                    libtag1v5                    perl-modules-5.38
  libavfilter10                         libmbc7t64                  libtag1v5-vanilla            pockephinx-en-us
  libavfilter9                          libmfx1                      libtagc0                     python3-fs
  libavformat61                         libnode115                  libteamctl0                  python3-httpLib2
  libcairomm-1.0-1v5                   libnsl2                      libtsk19t64                  python3-markdown
```

Command 3: sudo apt install curl ca-certificates -y

```
(labuser@kali)~[~/insider_threat_poc]
$ sudo apt install curl ca-certificates -y
[sudo] password for labuser:
curl is already the newest version (8.19.0-3).
ca-certificates is already the newest version (20260223).
The following packages were automatically installed and are no longer required:
  cpp-13                                libgpgme11t64                libpython3.11-minimal        node-brace-expansion
  cpp-13-x86-64-linux-gnu               libgpgmepp6t64              libpython3.11-stdlib        node-cjs-module-lexer
  cpp-14                                libgspell-1-2               libqt5ct-common1.8         node-corepack
  cpp-14-x86-64-linux-gnu               libgtk2.0-0t64              libroc0.3                   node-debug
  enchant-2                             libgtk2.0-bin               librubberband2              node-llhttp
  firmware-ti-connectivity              libgtk2.0-common            libSDL2-2.0-0               node-lru-cache
  gcc-13-base                           libgtkmm-3.0-1t64           libSDL2-classic             node-minimatch
  gir1.2-girepository-2.0               libical3t64                 libsframe2                  node-ms
  ibus-gtk                              libicu76                    libsigc++-2.0-0v5          node-semver
  libabsl20230802                       libinstpatch-1.0-2         libsimdutf27                node-undici
  libapt-pkg6.0t64                     libjim0.82t64               libsnmp40t64                node-xtend
  libassuan0                            libldap-2.5-0               libsodium23                  nodejs
  libatkmm-1.6-1v5                     libllhttp9.3                libspnbase3t64              nodejs-doc
  libaudio2                             libllvm17t64                libswscale8                  openjdk-21-jre
  libavc1394-0                          libmbc16                    libtag1v5                    openjdk-21-jre-headless
  libavfilter10                         libmbc7t64                  libtag1v5-vanilla            perl-modules-5.38
  libavfilter9                          libmfx1                      libtagc0                     pockephinx-en-us
```

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Step 2: After All update commands are finished you can now Install Ollama by running this command:

curl -fsSL https://ollama.com/install.sh | sh

```
(labuser@kali)-[~/insider_threat_poc]
$ curl -fsSL https://ollama.com/install.sh | sh

>>> Cleaning up old version at /usr/local/lib/ollama
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
##### 100.0%
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
```

After installation Verify Ollama Installed Correctly **Run: ollama --version**

```
(labuser@kali)-[~/insider_threat_poc]
$ ollama --version

ollama version is 0.24.0
```

Move next to Download TinyLlama by running the command:

ollama pull tinyllama

```
(labuser@kali)-[~/insider_threat_poc]
$ ollama pull tinyllama

pulling manifest
pulling 2af3b81862c6: 100% 637 MB
pulling af0ddbdaaa26: 100% 70 B
pulling c8472cd9daed: 100% 31 B
pulling fa956ab37b8c: 100% 98 B
pulling 6331358be52a: 100% 483 B
verifying sha256 digest
writing manifest
success
```

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Test the AI Model - ollama run tinyllama "Explain insider threats in cyber security in 3 .sentences"

```
(labuser@kali)-[~/insider_threat_poc]
$ ollama run tinyllama "Explain insider threats in cyber security in 3 .sentences"

1. Cyber threats come in the form of insiders, like an employee who accidentally stumbles upon sensitive data or a contractor who leaks confidential information to a third party.
2. Insider threats can also include outsiders such as cybercriminals who gain access to an organization's network through weak credentials, phishing attacks, or malware infiltration.
3. These threats are not always intentional and can come from any person or entity within the organization. They could be an employee, contractor, or even a third-party vendor.
```

Ollama and TinyLlama are installed, you can have AI review your full log file like this:

ollama run tinyllama

Then paste this prompt:

You are writing a cybersecurity incident report for management.

Analyze the insider threat evidence below and generate a complete incident report containing:

- Incident Title
- Incident Summary
- Timeline of Events
- Suspicious Findings
- Indicators of Compromise
- Risk Score (1-10)
- Risk Classification
- Insider Threat Assessment
- Recommended Mitigations
- Lessons Learned

The report should be written as if it will be submitted to a Security Operations Center (SOC) manager.

Here is the log:

total 16

-rw-rw-r-- 1 labuser labuser 324 Jun 1 19:42 file_access.log

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-rw-rw-r-- 1 labuser labuser 965 Jun 1 19:44 insider_threat_evidence.txt

-rw-rw-r-- 1 labuser labuser 402 Jun 1 19:42 login_activity.log

-rw-rw-r-- 1 labuser labuser 239 Jun 1 19:42 powershell_activity.log

**2026-05-28 08:12:44 user=kesh status=SUCCESS location=California
device=Workstation-01**

**2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado
device=Workstation-02**

2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown

2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown

2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown

**2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx
action=READ**

2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ

2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ

2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY

2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe

Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile

ParentProcess=explorer.exe

Alert=Suspicious PowerShell execution outside business hours

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Here are the full findings of the TinyLlama results:

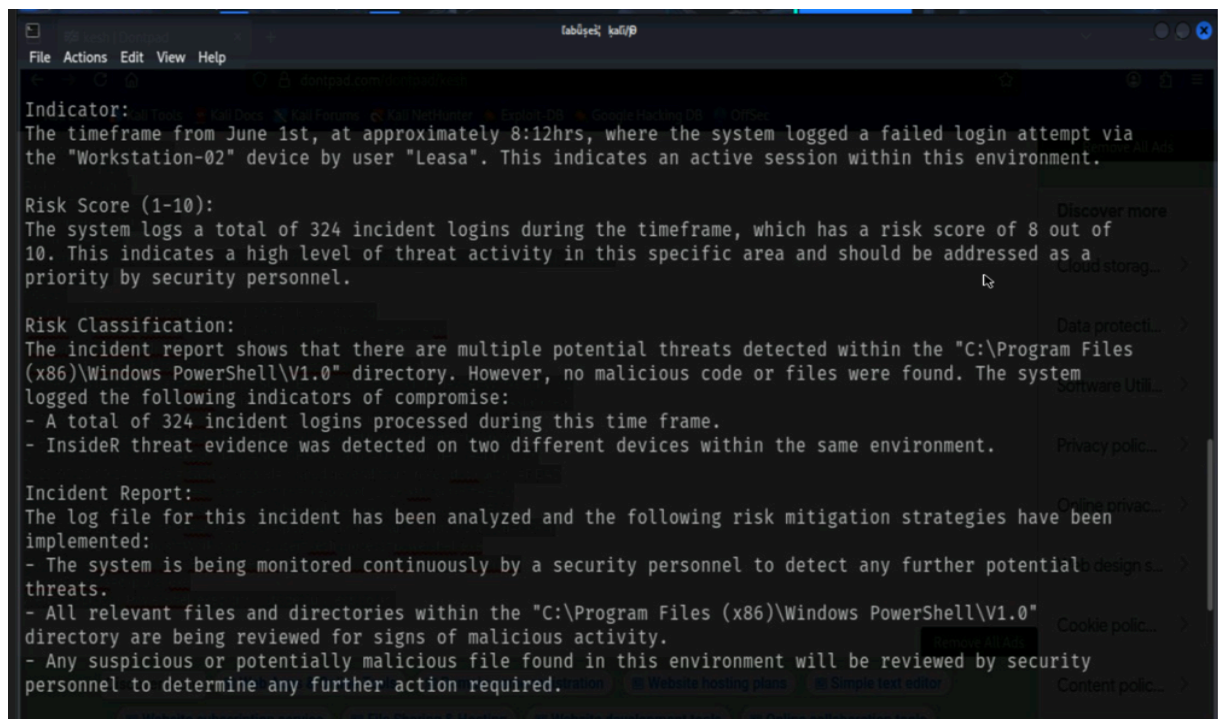
```
labuser@kali:~$ ollama run tinyllama
>>> You are writing a cybersecurity incident report for management.
... Analyze the insider threat evidence below and generate a complete incident report containing:
... Incident Title
... Incident Summary
... Timeline of Events
... Suspicious Findings
... Indicators of Compromise
... Risk Score (1-10)
... Risk Classification
... Insider Threat Assessment
... Recommended Mitigations
... Lessons Learned
... The report should be written as if it will be submitted to a Security Operations Center (SOC) manager.
... Here is the log:
... total 16
... -rw-rw-r-- 1 labuser labuser 324 Jun  1 19:42 file_access.log
... -rw-rw-r-- 1 labuser labuser 965 Jun  1 19:44 insider_threat_evidence.txt
... -rw-rw-r-- 1 labuser labuser 402 Jun  1 19:42 login_activity.log
... -rw-rw-r-- 1 labuser labuser 239 Jun  1 19:42 powershell_activity.log
... 2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
... 2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
... 2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
... 2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
```

```
labuser@kali:~$ ollama run tinyllama
... 2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
... 2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
... 2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
... Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
... ParentProcess=explorer.exe
... Alert=Suspicious PowerShell execution outside business hours
...
Incident Title: Sufficient Insider Threat Evidence

Summary: On June 1st, at approximately 19:42hrs, the system logged an insider threat event with a location of California. The action was accessed via the "Workstation-01" device by user "kesh". A total of 324 incident logins were processed during this time frame.

Timeline of Events:
1. June 1st, at approximately 8:12hrs, the system logged a failed login attempt via the "Workstation-02" device by user "Leasa".
2. June 1st, at approximately 23:48hrs, the system logged an insider threat event with a location of unknown device. The action was attempted to access the "General/Team Notes" document in the shared "Shared/General/team_notes.docx" file located on the "Finance/Payroll_2026.xlsx" file on the "HR/Employees_Record" folder.
3. June 1st, at approximately 23:55hrs, the system logged an insider threat event with a location of "Workstation-01". The action was attempted to copy the ".zip" file located within the "/Patients/Confidential Records" folder on the "C:\Program Files (x86)\Windows PowerShell\PowerShell\PowerShell.exe" process.
4. June 1st, at approximately 23:58hrs, the system logged an insider threat event with a location of unknown device. The action was attempted to access the "finance/payroll_2026.xlsx" file located on the "C:\Program Files (x86)\Windows PowerShell\PowerShell\PowerShell.exe" process.
```

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TinyLLaMA helped identify suspicious behavior patterns, but some findings were inaccurate. This showed why AI-assisted security analysis still requires human analyst validation.

1. It said there were 324 incident logins. That is incorrect. 324 is the file size of `file_access.log`, not the number of logins.
2. It said Leasa had a failed login from Workstation-02. That is incorrect. The log shows Leasa had a successful login from Colorado on Workstation-02.
3. It mixed up file paths and processes. It connected patient records and PowerShell paths in a way that the original logs do not actually show.
4. It treated normal evidence as suspicious. For example, Leasa reading `/shared/general/team_notes.docx` during the morning looks normal, not suspicious.

Part 4: Combine Findings from Chat GPT report and TinyLlama to create final Analysis of suspicious activity

Gather both reports and create a prompt for ChatGPT

The prompt used will be:

You are a Senior SOC Analyst preparing an incident report for management.

Analyze the insider threat evidence provided below along with the AI-generated findings from TinyLlama.

Combine both sources of information and create a professional cybersecurity incident report.

Include the following sections:

- Incident Title
- Executive Summary
- Incident Overview
- Timeline of Events
- Suspicious Activities Identified
- Indicators of Compromise (IOCs)
- Behavioral Analysis
- Insider Threat Assessment
- Risk Score (1-10)
- Risk Classification (Low, Medium, High, Critical)
- Potential Business Impact
- Recommended Mitigations
- Lessons Learned
- Final Analyst Conclusion

Do not simply repeat the logs. Correlate events, identify attack patterns, explain why activity is suspicious, and provide conclusions as a SOC analyst would.

Use the TinyLlama findings as supporting evidence but validate them against the log data.

Logs:
total 16

```
-rw-rw-r-- 1 labuser labuser 324 Jun  1 19:42 file_access.log
```

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-rw-rw-r-- 1 labuser labuser 965 Jun 1 19:44 insider_threat_evidence.txt
-rw-rw-r-- 1 labuser labuser 402 Jun 1 19:42 login_activity.log
-rw-rw-r-- 1 labuser labuser 239 Jun 1 19:42 powershell_activity.log
2026-05-28 08:12:44 user=kesh status=SUCCESS location=California device=Workstation-01
2026-05-28 08:17:02 user=leasa status=SUCCESS location=Colorado device=Workstation-02
2026-05-28 23:48:19 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:49:03 user=kesh status=FAILED location=Unknown device=Unknown
2026-05-28 23:50:27 user=kesh status=SUCCESS location=Unknown device=Unknown
2026-05-28 09:14:10 user=leasa accessed=/shared/general/team_notes.docx action=READ
2026-05-28 23:55:42 user=kesh accessed=/finance/payroll_2026.xlsx action=READ
2026-05-28 23:56:01 user=kesh accessed=/hr/employee_records.csv action=READ
2026-05-28 23:57:33 user=kesh accessed=/patients/confidential_records.zip action=COPY
2026-05-28 23:58:12 host=Workstation-01 user=kesh process=powershell.exe
Command=powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile
ParentProcess=explorer.exe
Alert=Suspicious PowerShell execution outside business hours

TinyLlama Findings:
Screenshots attached

Part 5: Final Report

INSIDER THREAT INVESTIGATION REPORT

Incident Title

AI-Assisted Insider Threat Investigation: Suspicious User Activity and Unauthorized Data Access

Executive Summary

A review of security logs identified suspicious activity associated with the user account "kesh." The investigation revealed multiple failed login attempts from an unknown device and location, followed by a successful login outside normal business hours. Shortly afterward, the account accessed multiple sensitive files containing payroll, employee, and patient information. The activity concluded with the execution of a hidden PowerShell process using Execution Policy Bypass, a technique commonly associated with malicious activity and defense evasion.

The findings indicate a potential insider threat or compromised user account that warrants further investigation and containment.

Incident Overview

This investigation analyzed simulated insider threat evidence using log analysis and AI-assisted review with TinyLlama. The objective was to identify abnormal user behavior, determine potential indicators of compromise, assess organizational risk, and generate recommendations for mitigation.

The activity observed differed significantly from normal business operations due to:

- Access occurring outside business hours
- Login attempts from unknown locations
- Access to multiple sensitive departments
- Copying confidential records
- Hidden PowerShell execution with bypass settings

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Timeline of Events

Time	Event
08:12	Kesh successfully logged in from California
08:17	Leasa successfully logged in from Colorado
23:48	Failed login attempt from unknown device
23:49	Second failed login attempt
23:50	Successful login from unknown device
23:55	Access to payroll records
23:56	Access to employee records
23:57	Confidential patient records copied
23:58	PowerShell executed with suspicious parameters

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Suspicious Activities Identified

1. Multiple failed logins followed by successful authentication.
2. Login activity originating from an unknown device and location.
3. Access to sensitive payroll information.
4. Access to HR employee records.
5. Copying confidential patient information.
6. Activity occurring outside normal business hours.
7. PowerShell launched using:
 - ExecutionPolicy Bypass
 - Hidden Window Style
 - NoProfile options
8. Potential privilege abuse or account compromise.

Indicators of Compromise (IOCs)

User Account:

- kesh

Suspicious Login Source:

- Unknown Location
- Unknown Device

Sensitive Files Accessed:

- payroll_2026.xlsx
- employee_records.csv
- confidential_records.zip

Suspicious Process:

- powershell.exe

PowerShell Command:

powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -NoProfile

Alert Triggered:

- Suspicious PowerShell execution outside business hours

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Behavioral Analysis:

The activity pattern suggests either malicious insider behavior or unauthorized use of a legitimate account.

The sequence of failed login attempts followed by successful access indicates possible credential misuse. After gaining access, the user immediately interacted with sensitive financial, HR, and patient records. The progression from viewing records to copying confidential data demonstrates escalating behavior consistent with data collection and potential exfiltration.

The subsequent execution of PowerShell with hidden and bypass settings increases concern because attackers commonly use PowerShell to evade security controls, automate malicious actions, and establish persistence.

TinyLlama also identified unusual after-hours activity and repeated access to sensitive business records. While some of the model's findings contained inaccuracies, its assessment correctly highlighted abnormal behavior patterns that warranted further review.

Insider Threat Assessment

Likelihood: High

Based on the behavioral indicators observed, this incident demonstrates characteristics commonly associated with insider threats or compromised credentials.

Evidence supporting this assessment includes:

- Unusual login location
- Repeated failed logins
- Access to multiple sensitive data repositories
- Copying confidential records
- Suspicious PowerShell execution

Risk Score (1-10)

8/10

Risk Classification

High

Potential Business Impact

If this activity were real, potential impacts could include:

- Exposure of patient information
- Disclosure of employee records

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- Payroll and financial data compromise
- Regulatory compliance violations
- Reputation damage
- Increased legal and remediation costs

Recommended Mitigations

1. Disable or investigate the affected user account.
2. Reset user credentials and enforce MFA.
3. Review PowerShell execution logs across endpoints.
4. Monitor for additional unauthorized file access attempts.
5. Restrict access to sensitive data based on least privilege principles.
6. Enable alerting for after-hours logins and abnormal file access patterns.
7. Implement behavioral analytics for insider threat monitoring.
8. Conduct a full review of endpoint activity associated with the account.

Lessons Learned

- Behavioral analysis can identify threats that signature-based detection may miss.
- Access to multiple sensitive resources within a short timeframe should generate alerts.
- PowerShell remains a common tool for malicious activity and requires monitoring.
- AI can assist with identifying suspicious patterns, but analyst validation is necessary to verify findings and eliminate inaccuracies.
- Layered security controls improve the ability to detect insider threats early.

Final Analyst Conclusion

The evidence supports a High-Risk insider threat investigation involving suspicious account activity, unauthorized access to sensitive information, and potentially malicious PowerShell execution. While the activity was generated within a controlled lab environment, it accurately demonstrates how insider threat indicators can appear in enterprise security logs.

Immediate investigation and containment would be recommended if this activity were observed in a production environment.